

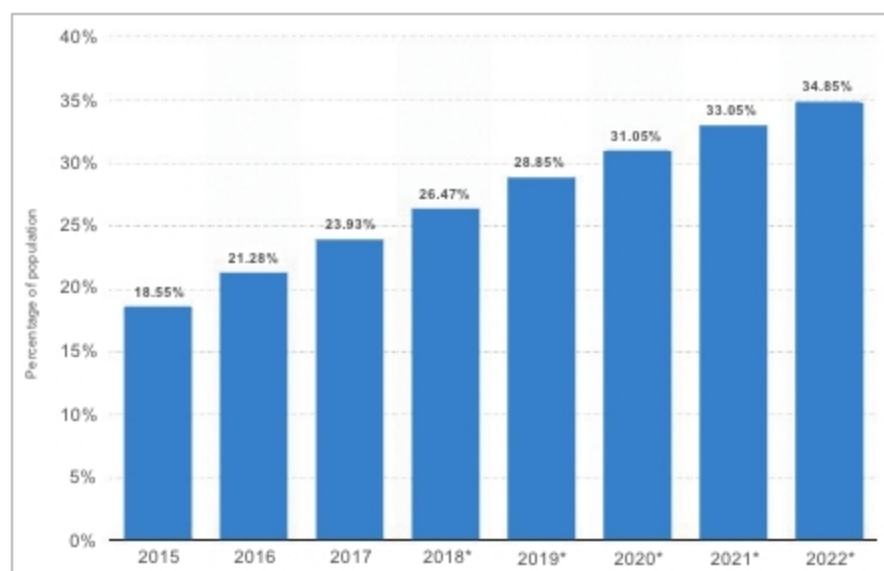


Figure 1: Prediction pattern of mobile phone Internet users (2015-2022)

Table 1: Prominent tools and libraries for network simulation

Software suite	Official Web based link
CISCO Packet Tracer	https://www.netacad.com/courses/packet-tracer
CNET	http://www.csse.uwa.edu.au/cnet/
CORE	https://www.nrl.navy.mil/itd/ncs/products/core
Cloonix	http://clownix.net/
DSA	http://www.iot-dsa.org/
GLOMOSIM	http://pcl.cs.ucla.edu/projects/glomosim
GNS-3	http://www.gns-3.net
IMUNES	https://github.com/imunes/

Please note that this list is indicative and not exhaustive. Other tools and libraries can be found on the Internet.

Key differences between 1G, 2G, 3G, 4G and 5G networks

1G: This mobile network refers to the first generation of cell phone technology for wireless communications. It focused on interactions in remote locations without any specific need for Internet and related services. 1G had analogue standards and protocols, so that at least voice based communication could be done from distant locations. The maximum speed expected from 1G communication was 2.4kbps with voice-only transmission, but there were huge packet drops or call drops with very little battery life.

2G: This network was a revolution in cell phone transmission at the time it came out, with messaging services integrated with the cellular technology. In this era, digital encryption techniques evolved so that overall communication could be made secure with a more effective spectrum of wireless transmissions. In this phase, the Short Message Service (SMS) and Multimedia Message Service (MMS) were integrated with traditional services for effective communications. There were speed constraints in 2G — up to 50kbps using General Packet Radio Service (GPRS) technology.

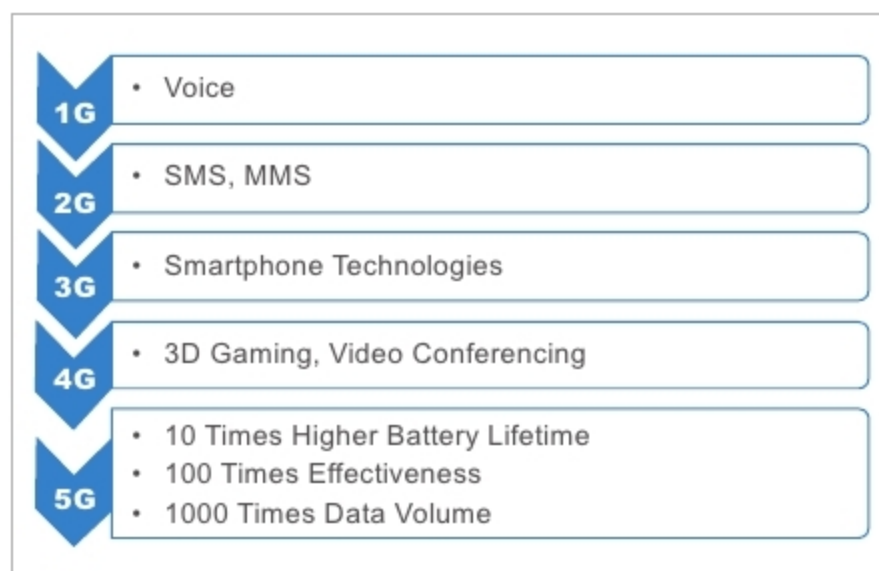


Figure 2: Cellular technologies

3G: Wireless communication using 3G technology was another revolution that integrated smartphone technologies with video downloading, e-mails and multimedia – at a speed of 2Mbps. The associated versions were 3.5G and 3.7G, with higher speeds for multiple corporate applications. 3G cellular technology offered high speed download packet access along with a higher degree of performance.

4G: Mobile telephone networks using 4G technologies provide advanced services including Internet TV, IP telephony, high definition live video conferencing, cloud services, high definition live gaming, 3D television and multimedia transmission at a speed of 100Mbps. 4G technology is today being used by most of the mobile network service providers in India as well as abroad.

5G: The technology associated with 5G cellular networks is quite sophisticated with a higher degree of accuracy and minimum delay. 5G networks are expected to have a tremendous speed of up to 10Gbps with very little downtime.

Simulating 5G networks using mmWave ns-3

The cellular network simulator of Millimetre Wave (mmWave) (<https://apps.nsnam.org/app/mmwave/>) provides the simulation patterns for 5G technologies so that upcoming network scenarios can be predicted. As 5G is the upcoming cellular technology, the behaviour of different network scenarios can be checked using such network simulators.

The mmWave module integrates the following features for 5G technologies, which can be used in simulated environments:

- Simulation of 5G networks with numerous nodes and base stations
- Cross-layered architecture of 5G networks
- Frequency spectrum of more than 6GHz
- Secured transmission
- Multi-layered connection with secured layer
- Support with LTE technologies so that a higher degree of performance can be achieved
- Multi-sector cellular networks
- Network carrier management and models
- Frequency spectrum management